



# Arithmetic Sequences

Sequences with a constant difference between consecutive terms have interesting properties and come up in many situations.

## Finding Common Differences

### Definitions

A sequence is an **arithmetic sequence** if the difference between consecutive terms is a constant.

$$d = a_n - a_{n-1}$$

The constant  $d$  is called the **common difference** of the sequence.

To find the common difference of a given sequence, select any term of the sequence and subtract it from the next consecutive term.

► **Think About It** If the differences between consecutive terms in a sequence are not equal, the sequence is not arithmetic.

### EXAMPLE 1

For the arithmetic sequence, find the common difference.

**A** 14, 11, 8, 5, 2, ...

#### SOLUTION

Find the difference between  $c_3$  and  $c_2$ .

$$d = 8 - 11 = -3$$

The common difference is  $-3$ .

**B**  $-9, -7.5, -6, -4.5, -3, \dots$

#### SOLUTION

Find the difference between  $h_2$  and  $h_1$ .

$$d = -7.5 - (-9) = -7.5 + 9 = 1.5$$

The common difference is 1.5. ■

## Writing and Using Recursive Rules for Arithmetic Sequences

### Recursive Rule for an Arithmetic Sequence

The formula for the common difference  $d = a_n - a_{n-1}$  can be rearranged to obtain the following recursive rule for arithmetic sequences:

$$\begin{array}{ccccc} \text{any term} & \leftarrow & a_n = a_{n-1} + d & \rightarrow & \text{common} \\ \text{(after first)} & & & & \text{difference} \\ & & \downarrow & & \\ & & \text{previous term} & & \end{array}$$

When you define an arithmetic sequence recursively, you also need to provide the value for  $a_1$ .

## EXAMPLE 2

For the sequence 12,  $-3$ ,  $-18$ ,  $-33$ ,  $-48$ ,  $\dots$ :

**A** Find a recursive rule.

### SOLUTION

Begin by finding the common difference.

$$d = -3 - 12 = -15$$

Substitute  $-15$  for  $d$  in the equation  $m_n = m_{n-1} + d$ . The recursive rule is  $m_n = m_{n-1} - 15$ .

The sequence starts with 12, so  $m_1 = 12$ .

**B** Use the recursive rule to find the next two terms in the sequence.

### SOLUTION

Find the next two terms,  $m_6$  and  $m_7$ .

$$\begin{aligned} m_6 &= m_{6-1} - 15 & m_7 &= m_{7-1} - 15 \\ &= m_5 - 15 & &= m_6 - 15 \\ &= -48 - 15 = -63 & &= -63 - 15 = -78 \end{aligned}$$

The next two terms of the sequence are  $-63$  and  $-78$ . ■

► **Think About It** Note that an infinite number of sequences can be defined by the recursive rule  $a_n = a_{n-1} - 15$ , such as  $-5$ ,  $-20$ ,  $-35$ ,  $-50$ ,  $\dots$ , and  $47$ ,  $32$ ,  $17$ ,  $2$ ,  $\dots$ , but only one has  $a_1 = 12$ .

# Writing and Using Iterative Rules for Arithmetic Sequences

Using the recursive rule, you can see an iterative rule for jumping straight to the  $n$ th term in an arithmetic sequence.

$$a_1 = a_1 + 0d$$

$$a_2 = a_1 + d = a_1 + 1d$$

$$a_3 = a_2 + d = (a_1 + d) + d = a_1 + 2d$$

$$a_4 = a_3 + d = (a_1 + 2d) + d = a_1 + 3d$$

... and so on. Notice that each multiple of  $d$  is always 1 less than  $n$ .

## Iterative Rule for an Arithmetic Sequence

The iterative rule for the  $n$ th term of an arithmetic sequence with first term  $a_1$  and common difference  $d$  is

$$a_n = a_1 + (n - 1)d.$$

This rule makes it possible to find any term of an arithmetic sequence if you know the first term and the common difference.

### EXAMPLE 3

For the sequence 16, 20, 24, 28, 32, ... :

**A** Find an iterative rule.

#### SOLUTION

The first term is  $f_1 = 16$ , and the common difference is  $d = 20 - 16 = 4$ . Substitute 16 for  $f_1$  and 4 for  $d$  in the rule  $f_n = f_1 + (n - 1)d$ , and then simplify.

$$\begin{aligned} f_n &= 16 + (n - 1)4 \\ &= 16 + 4n - 4 \\ &= 12 + 4n \end{aligned}$$

The iterative rule is  $f_n = 12 + 4n$ .

**B** Use the iterative rule to find the 82nd term in the sequence.

**SOLUTION**

To find the 82nd term, substitute 82 for  $n$  in the rule.

$$f_{82} = 12 + 4 \cdot 82 = 340$$

The 82nd term of the sequence is 340. ■

## Graphing Arithmetic Sequences

### EXAMPLE 4

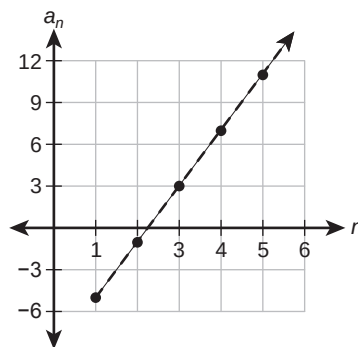
Graph the arithmetic sequence in the coordinate plane.

**A**  $a_n = -5 + 4(n - 1)$

**SOLUTION**

The domain is the set of positive integers. Make a table of some points and plot those points to get a partial graph.

| $n$ | $a_n = -5 + 4(n - 1)$                                     | $a_n$ |
|-----|-----------------------------------------------------------|-------|
| 1   | $a_1 = -5 + 4(\mathbf{1} - 1) = -5 + 4 \cdot 0 = -5 + 0$  | -5    |
| 2   | $a_2 = -5 + 4(\mathbf{2} - 1) = -5 + 4 \cdot 1 = -5 + 4$  | -1    |
| 3   | $a_3 = -5 + 4(\mathbf{3} - 1) = -5 + 4 \cdot 2 = -5 + 8$  | 3     |
| 4   | $a_4 = -5 + 4(\mathbf{4} - 1) = -5 + 4 \cdot 3 = -5 + 12$ | 7     |
| 5   | $a_5 = -5 + 4(\mathbf{5} - 1) = -5 + 4 \cdot 4 = -5 + 16$ | 11    |



► **Think About It** The graph of any arithmetic sequence will consist of discrete points that lie on a line.

**B**  $q_n = 12 + (-2)(n - 1)$

**SOLUTION**

Find the first five terms.

| $n$ | $q_n = 12 + (-2)(n - 1)$                                  | $q_n$ |
|-----|-----------------------------------------------------------|-------|
| 1   | $q_1 = 12 + (-2)(\mathbf{1} - 1) = 12 + (-2)(0) = 12 + 0$ | 12    |
| 2   | $q_2 = 12 + (-2)(\mathbf{2} - 1) = 12 + (-2)(1) = 12 - 2$ | 10    |
| 3   | $q_3 = 12 + (-2)(\mathbf{3} - 1) = 12 + (-2)(2) = 12 - 4$ | 8     |
| 4   | $q_4 = 12 + (-2)(\mathbf{4} - 1) = 12 + (-2)(3) = 12 - 6$ | 6     |
| 5   | $q_5 = 12 + (-2)(\mathbf{5} - 1) = 12 + (-2)(4) = 12 - 8$ | 4     |

